Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. The dataset was split into 80 percent training data and the rest for testing.

2 Results

Figure 1 shows the validation accuracy by dataset.

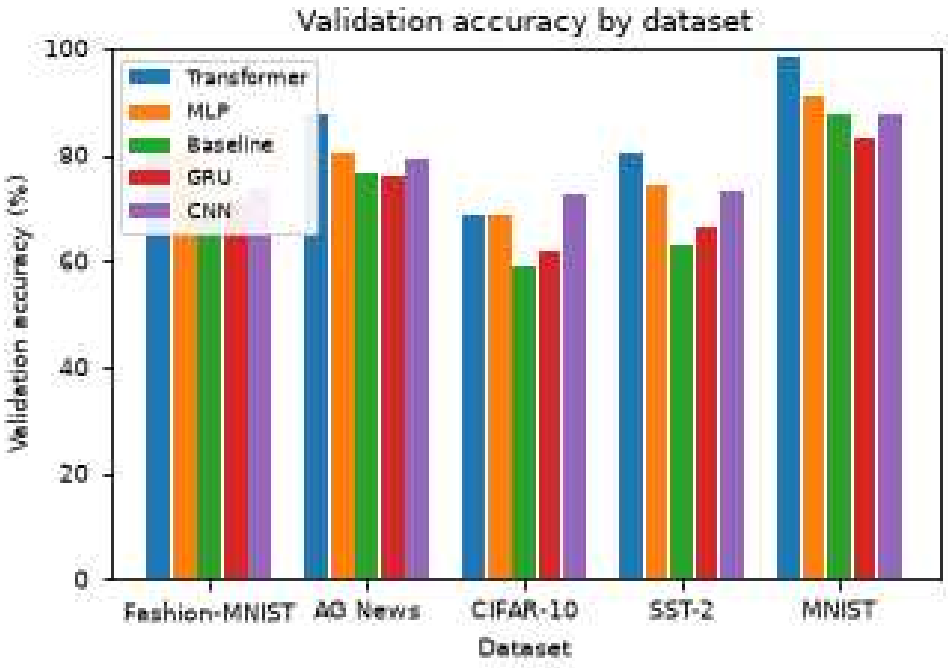


Figure 1: Validation accuracy by dataset for the models in this experiment.

Baseline achieves 82.5% validation accuracy on MNIST (Figure 1). As shown in Figure 1, CNN reaches 78.0% validation accuracy on SST-2. GRU achieves over 70% validation accuracy on AG News. CNN reaches 79.2% validation accuracy on AG News (Figure 1).

Figure 2 shows the training loss.

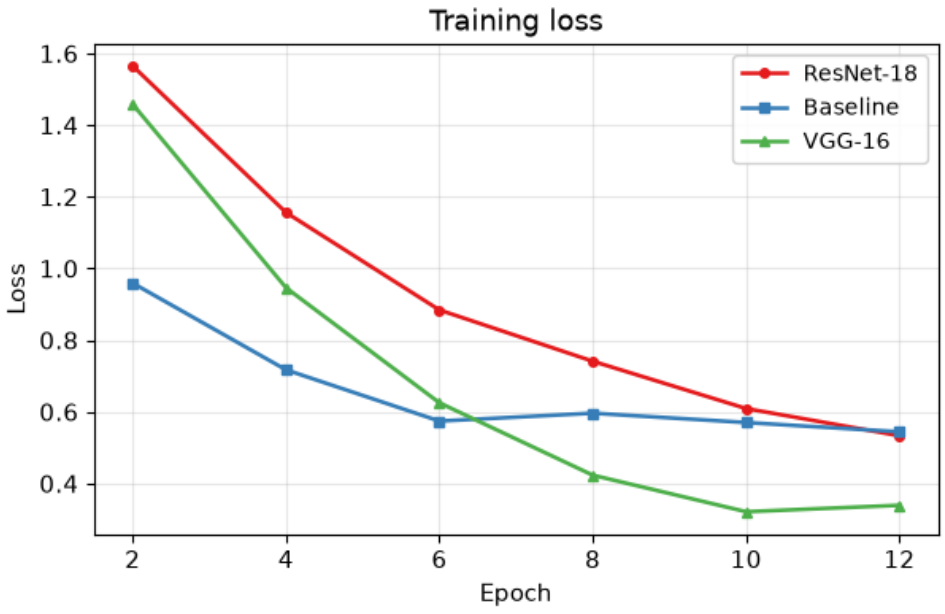


Figure 2: Training loss for the models in this experiment.

At epoch 6, the loss of Baseline is 0.601. At epoch 10, the loss of ResNet-18 is -0.075.

Figure 3 shows the accuracy by dataset.

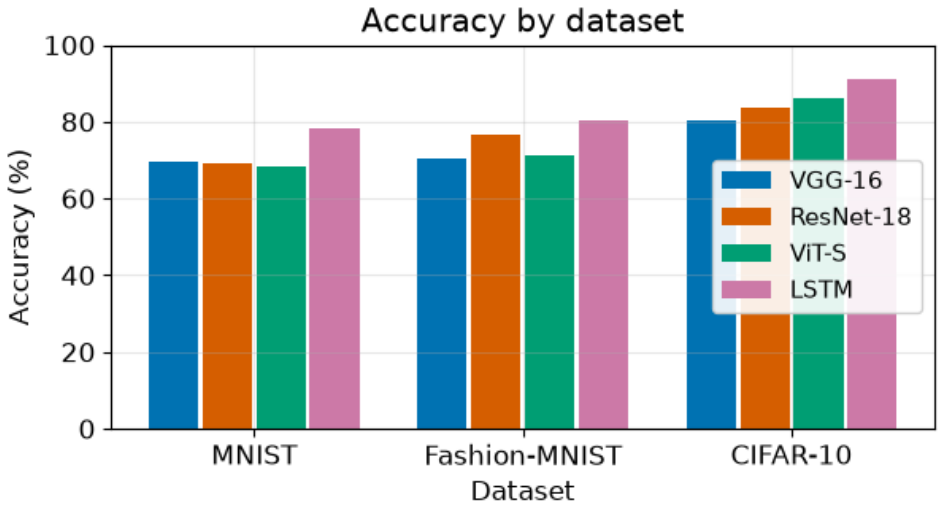


Figure 3: Accuracy by dataset for the models in this experiment.

ViT-S achieves 118.2% accuracy on MNIST. VGG-16 achieves 70.3% accuracy on Fashion-MNIST.

Figure 4 shows the f1 score of Ours by dataset.

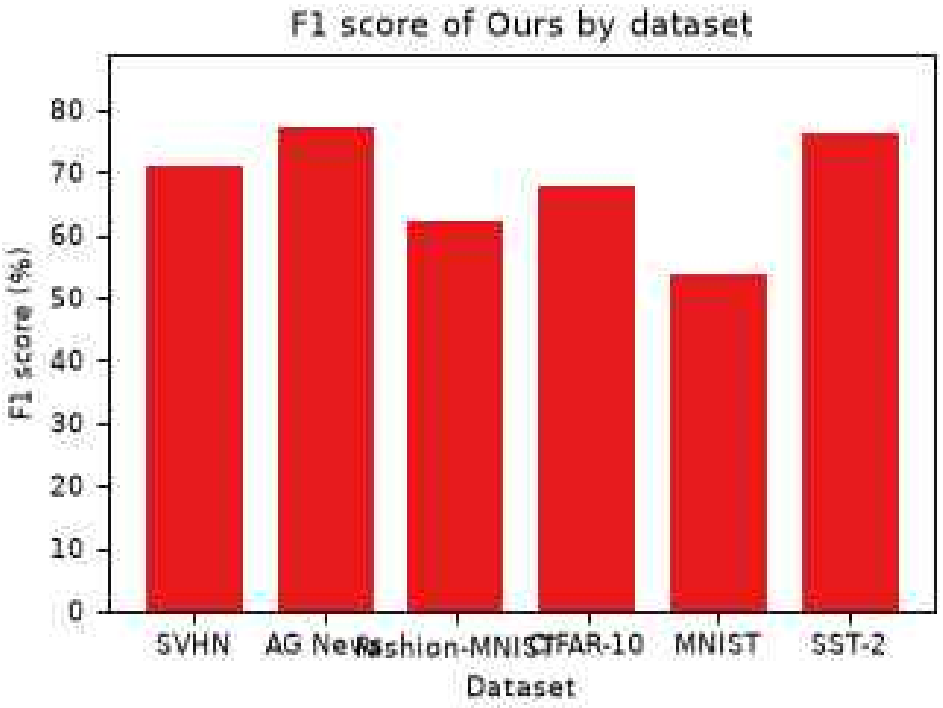


Figure 4: F1 score of Ours by dataset for the models in this experiment.

As shown in Figure 4, Ours achieves 58.5% F1 score on SST-2. Ours scores 13.9 points higher on SST-2 than on SVHN. On SST-2, Ours obtains a F1 score of 74.4%.

3 Discussion

Random Forest reaches 81.7% accuracy on IMDB. The differences between the models are discussed in the next section.